

Interpreting the age coefficient

1

Weak grip by age: summary output

```
klosa <- read_csv("klosa_men_grip_45plus.csv")
```

```
mod1 <- glm(weak_grip ~ age, family = binomial, data = klosa)
summary(mod1)
```

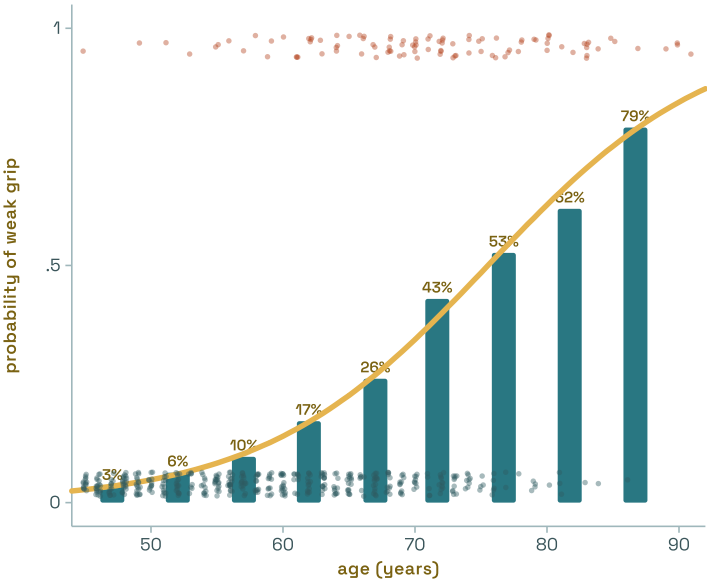
```
#>               Estimate Std. Error z value Pr(>|z|)
#> (Intercept) -8.82431    0.31925   -27.64  <2e-16 ***
#> age          0.11683     0.00476    24.57  <2e-16 ***
```

- Each **dot** is one man in the data, at **1** (weak grip) or **0** (not weak).
- The **gold curve** is the fitted model. The change in probability per year is **not the same everywhere**: small at young ages, larger in the middle, and small again at old ages.

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \text{ age})}}$$

$$\hat{p} = \frac{1}{1 + e^{-(-8.824 + 0.1168 \text{ age})}}$$

- The summary gives **one number** for age, $\beta_1 = 0.1168$. It is the change in the **log-odds** for each extra year, not a change in probability. The next panel shows what that means.



2

The same model on two scales

- Below we show the derivation of the log-odds form from the probability form of the logistic equation.

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \text{ age})}}$$

Let's let η (eta) be the linear part in the exponent:

$$\eta = \beta_0 + \beta_1 \text{ age}$$

$$p = \frac{1}{1 + e^{-\eta}}$$

Subtract p from 1:

$$1 - p = \frac{1 + e^{-\eta}}{1 + e^{-\eta}} - \frac{1}{1 + e^{-\eta}} = \frac{e^{-\eta}}{1 + e^{-\eta}}$$

We now have two identities for p and $1 - p$. Substitute them into $p/(1 - p)$ (to divide by a fraction, flip it and multiply):

$$\frac{p}{1 - p} = \frac{1}{1 + e^{-\eta}} \div \frac{e^{-\eta}}{1 + e^{-\eta}}$$

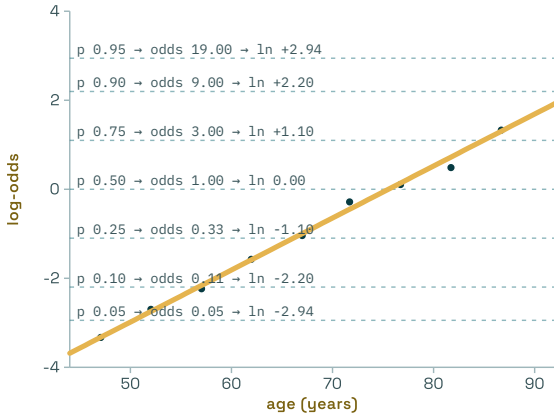
$$= \frac{1}{1 + e^{-\eta}} \cdot \frac{1 + e^{-\eta}}{e^{-\eta}} = \frac{1}{e^{-\eta}} = e^{\eta}$$

Take the natural log of both sides:

$$\ln \frac{p}{1 - p} = \ln(e^{\eta}) = \beta_0 + \beta_1 \text{ age}$$

$$\ln \frac{p}{1 - p} = \beta_0 + \beta_1 \text{ age} = -8.824 + 0.1168 \text{ age}$$

- The age effect is the **slope** of that line: each extra year adds **+0.1168** to the log-odds. That is β_1 from the summary.



3

Odds ratios are comparisons

CONTINUOUS PREDICTOR: AGE

```
mod2 <- glm(weak_grip ~ age,
  family = binomial, data = klosa)
broom::tidy(mod2, exponentiate = TRUE, conf.int = TRUE)
```

```
#> term      estimate std.error statistic p.value  conf.low conf.high
#> (Intercept) 0.000147   0.319    -27.6    <2e-16  7.83e-5  2.73e-4
#> age         1.124     0.00476    24.6    <2e-16  1.114   1.135
```

$$e^{0.1168} = 1.124$$

A **one-year difference in age** is associated with **1.124 times the odds**, or **12.4% higher odds**, of weak grip.

Report: “Each additional year of age was associated with 12% higher odds of weak grip (OR 1.12, 95% CI 1.11–1.13, $p < 0.001$).”

ODDS RATIO → PLAIN WORDS

0.25

75% lower odds
¼ the odds

0.5

50% lower odds
half the odds

0.8

20% lower odds

1.0

no change

1.12

12% higher odds
(age, per year)

1.5

50% higher odds

2.0

100% higher
twice the odds

4.0

4 times the odds

Polytomous predictor: age group

SAME KLOSA DATA, THREE LEVELS

```
klosa <- klosa |>
  mutate(age_group = cut(age,
    breaks = c(45, 65, 75, Inf),
    right = FALSE))
```

```
mod3 <- glm(weak_grip ~ age_group,
  family = binomial, data = klosa)
```

- R uses the first level, **45–64 years**, as the reference. With three levels, the model gives **two coefficients** and **two odds ratios**.

```
broom::tidy(mod3, exponentiate = TRUE, conf.int = TRUE)
```

```
#> term      estimate std.error statistic p.value  conf.low conf.high
#> (Intercept) 0.0958    0.0693   -33.8    <2e-16  0.0834   0.110
#> age_group65-74  5.22     0.0936    17.7    <2e-16  4.35     6.28
#> age_group75+  14.4     0.119     22.5    <2e-16  11.5    18.3
```

65–74 vs 45–64: 5.22 times the odds.

75+ vs 45–64: 14.44 times the odds.

Report: “Compared with men aged 45–64, men aged 65–74 had 5.22 times the odds of weak grip (95% CI 4.35–6.28), while men aged 75+ had 14.44 times the odds (95% CI 11.44–18.22).”